

Scalability & Future Plan

Date	5 July 2026
Team ID	
Project Name	EduGenie
Maximum Marks	1 Mark

Scalability & Future Plan:

The following section outlines the scalability strategy and future development roadmap for the EduGenie platform. It highlights the current system limitations, proposed improvements to support increased users and enhanced performance, and planned future features aimed at expanding the platform's capabilities and delivering a more advanced AI-powered learning experience.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	JSON file-based database.	Limits concurrent writes and database performance at scale.	High
2	Single-process local server.	Limits handling of large concurrent user traffic.	Medium
3	Local file uploads.	Can fill server storage quickly with high upload volume.	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports single-digit concurrent users locally.	Transition server deployment to an ASGI production server using Gunicorn/Uvicorn workers.
2	Data Storage	Flat JSON files in local directory.	Migrate database to PostgreSQL or MongoDB with proper ORM schema indexing.
3	Performance	Direct API calling with no caching.	Implement Redis caching for frequent educational queries and concept explanations.
4	Security	JWT stored in HTTPOnly cookies, local hashing.	Integrate secure OAuth2 providers, SSL encryption, and API rate limiting.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
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Phase 2	Migrate JSON data store to PostgreSQL database and setup SQLAlchemy ORM.	July-Aug 2026	Enables robust, multi-user concurrency and database normalization.
Phase 3	Add multi-language translation support for educational responses.	Aug-Sep 2026	Expands accessibility for students in non-English speaking regions.
Phase 4	Integrate speech-to-text input and text-to-speech reading directly into UI.	Sep-Oct 2026	Enhances usability for visually impaired and auditory learners.